# Strategy-Supervised Autonomous Laparoscopic Camera Control via Event-Driven Graph Mining

Keyu Zhou, Yahao Wu, and Shunlei Li*

***Abstract*—Despite significant advancements in robotic surgery, bridging high-level strategic reasoning and precise low-level motion control remains a critical challenge for achieving expert-level autonomy. We propose a strategy-grounded framework that couples high-level vision-language inference with low-level IBVS-RCM control. By mining reusable strategy primitives from event graphs for structured supervision, the proposed system reduces centering error by 35.26% and image shaking by 62.33% compared with manual human control in *ex vivo* trials.**

***Index Terms*—Robotic laparoscopy, autonomous camera control, vision-language models, strategy mining, graph clustering, human-in-the-loop robotics.**

## I. Introduction

Minimally Invasive Surgery (MIS) has revolutionized clinical interventions, where the laparoscopic camera serves as the surgeon's primary sensory interface. Maintaining an optimal and stable Field-of-View (FoV) is critical for surgical safety, yet traditional manual control by assistants is often hindered by physiological fatigue and miscommunication [1]. While automation via Image-Based Visual Servoing (IBVS) has been explored, these systems remain predominantly reactive and lack the semantic depth required to anticipate complex surgical workflows [2]. Furthermore, general-purpose Vision-Language Models (VLMs) struggle with the specialized logic and stringent safety requirements inherent in the operating room [3].

To address this gap, we propose a strategy-grounded hierarchical framework that decouples high-level reasoning from safety-constrained execution [4]. Our approach explicitly transforms expert intuition into structured control signals. The system operates in two core stages:

- Offline Strategy Discovery: Raw surgical demonstrations are parsed into a taxonomy of interaction, depth, and view-quality events. These events are organized into attributed event graphs to mine 12 reusable strategy primitives, capturing tacit handling patterns of expert surgeons [5].
- Online Strategy Control: A fine-tuned VLM processes real-time visual context and optional speech input to predict motion intentions. These intentions are executed by an IBVS-RCM controller that computes motion magnitudes while strictly enforcing Remote Center of Motion (RCM) [6] constraints to ensure safe trocar-point pivoting.

By anchoring autonomous control in expert-mined strategies, our framework provides a robust and interpretable assistance paradigm. The overall pipeline consists of three stages: offline parsing, strategy discovery, and online control, as illustrated in Fig. 1 [4].

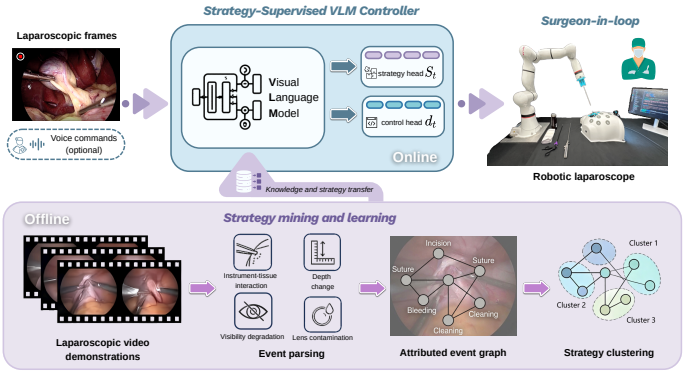


**Fig. 1:** The proposed framework integrates offline graph-mined strategy primitives with an online VLM-based policy and safety-constrained IBVS-RCM control to achieve stable, context-aware autonomous laparoscopic camera guidance.

## II. Methods

Offline Event Parsing: Videos are parsed into three event categories [7]. *Interaction events* are detected using a deformation score $S_{\text{def}}(t)$, defined as the ratio between motion magnitude in the tool-centric region and that in a distal reference region:

$$S_{\text{def}}(t) = \frac{\text{median}_{R_{\text{near}}(t)} \|\mathbf{u}\|_2}{\text{median}_{R_{\text{ref}}(t)} \|\mathbf{u}\|_2 + \epsilon} \tag{1}$$

where $R_{\text{near}}$ and $R_{\text{ref}}$ denote the tool-centric and distal reference regions, respectively; a larger score indicates stronger local motion around the tool relative to the background. These events are encoded as attribute vectors $\mathbf{x}_k = [\mathbf{x}_k^{\text{state}} \| \mathbf{x}_k^{\text{action}}]$ [8].

Strategy Discovery: Events form an attributed graph $(A, X, S)$ that encodes temporal and semantic affinity [5]. We apply *Weighted Symmetric Boosted Graph Clustering* (WSBGC), combining a graph attention autoencoder with symmetric NMF, to mine $K = 12$ strategy primitives (e.g., *re-centering* and *withdrawal*), which provide structured supervision for policy learning [5].

Real-Time Control: A fine-tuned VLM (Qwen2.5-VL) predicts the strategy $\hat{s}_t$ and 3-DoF motion directions $\hat{\mathbf{d}}_t$ from images and speech [3]. The training objective $\mathcal{L}$ combines a

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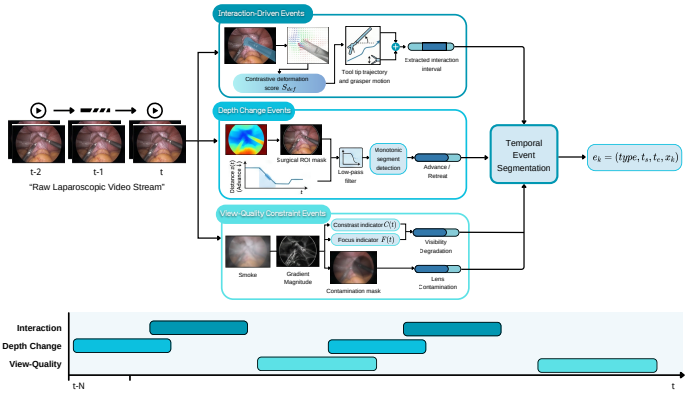


**Fig. 2:** Three-branch parsing fused into structured events $e_k$ and visualized over time.

directional classification loss with an auxiliary cross-entropy loss for strategy supervision:

$$\mathcal{L} = \mathcal{L}_{\text{dir}} + \lambda_s \mathcal{CE}(s_t^*, \hat{s}_t) \tag{2}$$

The predicted intentions $\hat{\mathbf{d}}_t$ are then executed by an IBVS-RCM controller that enforces RCM and safety constraints at the trocar entry point [6]. This design enables context-aware and mechanically safe camera guidance [9].

## III. Experiments and Results

Implementation and Setup: The framework was trained on a dataset of 109 laparoscopic cases (1.17M samples) using a Qwen2.5-VL (7B) backbone with LoRA fine-tuning. Validation was conducted on a ROKAE ER7Pro robotic platform. The setup included a synthetic abdominal cavity and *ex vivo* porcine tissues (intestine and stomach) for stitching and dissection tasks. The pipeline integrates VLM inference at 10 Hz with an IBVS-RCM controller at 100 Hz.

**TABLE I:** Comparison of Field-of-View Centering (FOV) and Image Shaking performance.

| Metric | Method | $1^{st}$ | $2^{nd}$ | $3^{rd}$ | $4^{th}$ | $5^{th}$ | $6^{th}$ | $7^{th}$ | $8^{th}$ | $9^{th}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| FOV | Human | 142.12 | 130.45 | 163.61 | 121.02 | 121.54 | 116.17 | 187.00 | 109.09 | 143.25 |
| | Ours | 62.47 | 83.52 | 104.01 | 97.35 | 94.07 | 99.04 | 77.59 | 66.86 | 114.20 |
| Image Shaking | Human | 1.45 | 1.68 | 1.86 | 1.68 | 1.07 | 1.81 | 1.36 | 1.63 | 1.48 |
| | Ours | 0.63 | 0.47 | 0.77 | 0.67 | 0.47 | 0.61 | 0.58 | 0.69 | 0.38 |

***Note:*** Values are in pixels. FOV denotes centering error, and Image Shaking denotes stabilization.

Key Results:

- Perception Performance: As shown in Table II, the offline event detection module achieved an average F1-score of 0.86 and a mean tIoU of 0.71. Strategy clustering ($K = 12$) also showed strong alignment with expert annotations, with a purity of 0.81.
- Control Efficacy: As shown in Table I, the proposed autonomous system achieved a 35.26% reduction in centering error ($e_{cen}$) and a 62.33% reduction in image shaking ($S$) compared with manual human control.
- Stability and Safety: The system maintained smooth trajectories, with a high-frequency spectral energy ratio of approximately $10^{-4}$.
- Autonomous Recovery: Severe fogging and lens contamination were mitigated through an automated "extraction-cleaning-restoration" sequence with an average recovery time of 25 s. In our test set, speech-based refinement achieved 100% interpretation accuracy.

These results indicate that the strategy-grounded VLM policy can effectively translate expert intuition into stable, precise, and safe robotic camera guidance.

**TABLE II:** Performance of Event Detection (tIoU $\geq 0.5$)

| Event | Precision | Recall | F1-score | Mean tIoU |
|---|---|---|---|---|
| Interaction | 0.86 | 0.81 | 0.83 | 0.67 |
| Depth deviation | 0.90 | 0.88 | 0.89 | 0.74 |
| Visibility degradation | 0.84 | 0.79 | 0.81 | 0.64 |
| Lens contamination | 0.93 | 0.91 | 0.92 | 0.80 |
| **Average** | **0.88** | **0.85** | **0.86** | **0.71** |

## IV. Conclusion

We proposed a hierarchical framework that bridges surgical semantics and robotic precision through graph-mined strategy discovery and VLM-based prediction. By translating expert intuition into robust IBVS-RCM control, the system achieved a 35.26% reduction in centering error and a 62.33% reduction in image shaking in *ex vivo* trials. Future work will focus on expanding the strategy library and conducting *in vivo* validation toward clinical integration.

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